Lesson 6: Research You Can Trust

Module 2 | 60 minutes | Sources and fact-checks | Adults (18+) · ages 18–65

Objective: Research a work question with AI search tools that show sources, open and check every source, spot made-up or old information, compare two tools, and write a fact-checked one-page note.

Success criteria

- I can ask an AI search tool for sources and official websites.
- I can open and check every source for the exact claim, publisher and date.
- I can spot hallucinations and old information and remove them.
- I can write a one-page note where every point has a checked source.

Key vocabulary

- **source / citation**: The web page or document a fact comes from, shown as a link or number.
- **official source**: The organisation that makes the rule or owns the facts, like a ministry, regulator or the company itself.
- **hallucination**: A made-up fact, number or link that the AI says confidently.
- **outdated information**: Facts that were true once but have changed, like an old fee or deadline.
- **fact-check**: Checking each claim against a trustworthy source before you use it.

You will need

- Projector or screen share with Kitabash and the AI tools for this lesson
- Devices: 1 per learner (laptop, tablet or phone)
- Learner accounts only where allowed: see the course safety notes
- Printed worksheet (optional)
- Project files: l06_research_note_kit.md, l06_teacher_demo_script.md

Timed lesson plan (60 min)

Time	Stage	What happens
5 min	**Warm-up game**	“Real or rumour?” team game (see Unplugged).
4 min	**Watch**	Play the lesson video. Pause at the 2019 blog: poll “Would you have clicked the source?”
7 min	**Live demo**	Perplexity with sources, open every link, the same question in Gemini, official site check, fact-check table in ChatGPT.
7 min	**Prompt Lab 1**	Prompt Lab: a research prompt with sources, for the learner’s own question. Run it in one tool.

6 min	**Quick competition**	“Can you rely on this source?” sort and “Spot the hallucination”, played as a team race.
10 min	**Breakout pairs: source detectives**	Pairs run the same question in a second tool, then open every source and fill the source-check table.
6 min	**Prompt Lab 2**	Prompt Lab: the fact-check table. Remove anything “could not verify”.
6 min	**Show-and-tell**	Three learners show one claim they removed and why.
5 min	**Probe**	The research routine (order) and New situation: two different dates (individual). Start the one-page note.
4 min	**Wrap-up**	Exit ticket and homework.

Unplugged warm-up

Real or rumour? (5 min): read out 6 short “facts” in the style of WhatsApp forwards and news lines (some true and general, some obviously made up, one true but old). Teams hold up “REAL”, “RUMOUR” or “CHECK!” cards. The winning habit is “CHECK!” for anything with a number and no source.

Teaching the idea

Adults often use AI as a search engine and copy whatever it says. This lesson changes that habit. Show AI search tools that give sources (Perplexity, Gemini, ChatGPT’s search option), then teach the journalist’s routine: open every source, find the exact sentence, check the publisher and the date, and remove anything you cannot verify. Compare two tools: if they disagree, the official source decides. Learners finish with a one-page market or policy note in which every point has a link and a date they checked themselves.

Analogy: AI search is like a helpful colleague who says “I read somewhere...”. Before you put it in a report, you ask, “Show me where,” and you read it yourself.

Live demo: 1) Open Perplexity and type the home-bakery question with “Give sources and prefer official websites.” 2) Click every numbered source; for each, read the sentence aloud and point to the date and publisher. 3) Ask Gemini the same question; compare in a two-column table on the board (agree / differ). 4) Open the official FSSAI website and show where to confirm current rules (do not read out fees or limits as facts; say “these change, check the site on the day”). 5) Ask ChatGPT for a fact-check table of four claims. Highlight any “could not verify” and remove it from the note.

Questions to ask

Ask	Listen for
What is the difference between an AI answer with sources and one without?	With sources you can open and check; without sources it is only a starting idea.

What three things do you check on every source?	That it says the exact claim, who published it, and the date.
Two tools disagree. What decides?	The official source.
Name two red flags of a hallucination.	A number with no source, a link that does not support the claim, a dead link, an old date.
What goes at the bottom of your note?	A line saying AI tools were used and all sources were opened and checked on a date.

Differentiation

Support
Give learners a ready question (home bakery registration) and a source-check table to fill (Claim, Link, Exact sentence, Publisher, Date, OK?).

Stretch
Research a market question for your city (for example, which festive products sell most) and find at least one official or primary data source; write what is still unknown.

Common misconceptions

Misconception	How to fix it
If there is a link, the fact is true.	Open the link; it may not say what the AI claims, or it may be old.
AI search always uses the latest information.	It may quote old pages. Check the date on every source.
The tool that sounds more confident is right.	Confidence means nothing; the official source decides.
Research with AI means no reading.	AI finds and drafts quickly; you still read the key sentences.

Assessment

Use the Prompt Lab results (auto-checked), the quick checks and the Probe answers. Written answers arrive in your review queue.

Exit ticket
Name one red flag that tells you an AI research answer may be wrong.

Homework
Finish your fact-checked one-page note on a real work question. Share it with a colleague and ask them to open two of your links.

Answer key — Kitabash activities

1. Prompt Lab: a research prompt with sources (perform · ai lab)

Checklist: Ask for sources or links · Ask for recent information · Prefer official websites · Set a format · What to do if unsure

Strong example:

```
I run a small home bakery in Pune. What food-safety registration do I need in India?
Answer in 5 bullet points, with a source link and date for each point, and prefer
official government websites. Tell me if any rule changed recently, and if you are not
sure about something, say so.
```

2. Prompt Lab: the fact-check table (perform · ai lab)

Checklist: Refer to the claims · Ask it to verify each one · Ask for a source link · Flag wrong or outdated claims · Ask for a table

Strong example:

```
Here are 4 claims from my research note. For each claim, verify it using official
sources, give the source link and its date, and mark it Supported, Not supported,
Outdated or Could not verify. Use a table with columns Claim, Status, Source, Date.
Claims: [paste claims]
```

3. Can you rely on this source for a work note? (perform · categorize)

Good source to rely on: An official government website ending in gov.in, A dated press release from a ministry or regulator, A company's own annual report (for facts about that company), A well-known newspaper article with a date and author
Check elsewhere first: A WhatsApp forward with no name or date, A blog post from 2018 about today's rules, An AI answer with no links, A video from an unknown channel promising "secret rules"

4. Spot the hallucination (perform · multiple choice)

Answer: The AI linked a source that does not support its claim, so treat the number as made up and remove it

5. The research routine (perform · ordering)

Order: Write a clear question and who it is for → Ask an AI search tool for sources and official sites → Ask a second tool the same question → Open every source and find the exact sentence and date → Remove anything you could not verify → Write the note with links, dates and an AI-use label

6. New situation: two different dates (probe · multiple choice)

Answer: Open the official scheme or government page and use the date it shows, with the page date noted

Answer key — worksheet

1. **Key words:** source / citation = The web page or document a fact comes from, shown as a link or number. · official source = The organisation that makes the rule or owns the facts, like a ministry, regulator or the company itself. · hallucination = A made-up fact, number or link that the AI says confidently. · outdated information = Facts that were true once but have changed, like an old fee or deadline.
2. **Prompt Lab: a research prompt with sources:** Include: Ask for sources or links; Ask for recent information; Prefer official websites; Set a format; What to do if unsure. Example: I run a small home bakery in Pune. What food-safety registration do I need in India? Answer in 5 bullet points, with a source link and date for each point, and prefer official government websites. Tell me if any rule changed recently, and if you are not sure about something, say so.
3. **Prompt Lab: the fact-check table:** Include: Refer to the claims; Ask it to verify each one; Ask for a source link; Flag wrong or outdated claims; Ask for a table. Example: Here are 4 claims from my research note. For each claim, verify it using official sources, give the source link and its date, and mark it Supported, Not supported, Outdated or Could not verify. Use a table with columns Claim, Status, Source, Date. Claims: [paste claims]
4. **Can you rely on this source for a work note?:** Good source to rely on: An official government website ending in gov.in, A dated press release from a ministry or regulator, A company's own annual report (for facts about that company), A well-known newspaper article with a date and author · Check elsewhere first: A WhatsApp forward with no name or date, A blog post from 2018 about today's rules, An AI answer with no links, A video from an unknown channel promising "secret rules"
5. **Spot the hallucination:** B. The AI linked a source that does not support its claim, so treat the number as made up and remove it — A link is only useful if it actually says what the AI claims. This number has no support, so it does not go in your note.

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